INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351

Soil and Water Engineering I

Lecture 2

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ANALYSIS OF METEOROLOGICAL DATA SUCH AS PRECIPITATION AND EVAPOTRANSPIRATION

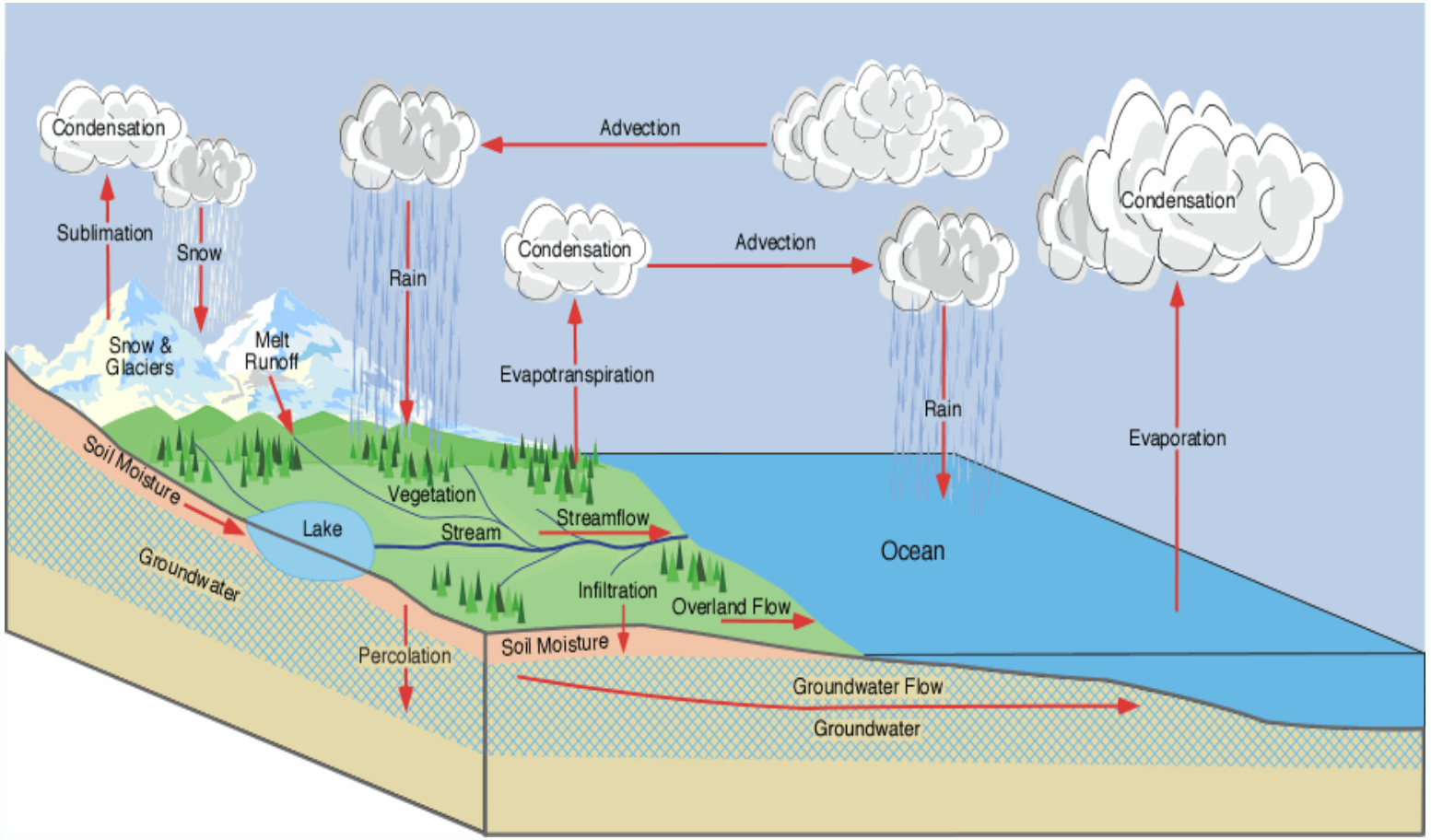


Fig. 1: The hydrological cycle

Some Meteorological/Climatic Data of Interest

- Humidity
- Temperature
- Radiation
- Wind (Speed and direction): It has the power law profile i.e. $\mathbf{U/U_o=(Z/Z_o)^{0.15}}$
- U_o is wind speed at height Z_o
- U is wind speed at some higher level Z
- Precipitation

Examples of Forms of Precipitation

• Drizzle/Mist	Ø 0.1-0.5 mm
• Virga	Evaporating streaks of water hanging in the air
• Rain	Liquid water Ø ≥0.5 mm
• Glaze	Ice coating from super-cooled water deposited by rain
• Rime	White, opaque deposits of ice granules
• Snow	Ice crystals
• Blizzard	Large amounts of fine, dry powdery particles of snow
• Hail/Hailstones	Conical, spheroidal or irregular
• Sleet	Transparent, globular solid, grains of ice Ø <0.5 mm.
• Dew	
• Fog	
• Ice	
• Condensation	

Occurrence of Precipitation/Importance in Agriculture

- Precipitation occurs by the cooling, condensation, nucleation and growth into rain drops. Water vapour in the atmosphere cools to its dew-point temperature, water vapour begins to condense as water droplets become visible as clouds. The dew point temperature is usually reached by lifting air into the colder, higher levels of the atmosphere.
- It is important to humankind as a source of drinking water and for agriculture. It cleanses the air and maintains the levels of lakes, rivers and oceans, which are sources of food and recreation.
- It determines the start and end of the growing season, soil moisture condition during the growing season, length of the growing season. It also determines the timing of farming activities like land preparation, sowing, fertilizer application, weed control.
- Rainfall is critical during the phenological stages of crops, availability of pasture for grazing animals, germination and establishment of seeds/seedlings.

Types of precipitation

- Relief or Orographic
- Convective
- Cyclonic or Frontal-Air moving into low pressure areas, displacing air in such areas upward to cool and precipitate. Remember that lapse rate is **6.5°C/1000 m**
- Artificial: Precipitation can also be induced artificially by seeding clouds

Processes in the formation of relief rainfall are as follows:

- Warm moist air is forced to rise over a mountain
- Warm air cools as it rises
- Condensation occurs
- Clouds form
- rain falls

Processes in the formation of Convectional rain are as follows:

- the sun heats the ground which heats the air/evaporates water from surfaces
- warm air rises
- air cools as it is forced to rise
- condensation
- clouds form
- rain falls

Processes in the formation of frontal rain

- mass of warm air meets a mass of colder air
- lighter warm air rises over heavier colder air
- warm air cools
- condensation
- clouds form
- rain falls

Climate data processing and analysis

- The internationally accepted convention, as recommended by the World Meteorological Organisation (WMO), is that a 30-year period is a basic climatic time scale.
- Climate statistics are, therefore, calculated for the consecutive 30-year time periods 1901-1930, 1931-1960 and 1961-90. These periods are known as **climatological standard normals**.
- Because the climate is changing rapidly nowadays, climate characteristics are sometimes re-calculated every 10 years for the period of the recent 30 years, i.e. 1961 - 1990, 1971 - 2000, although the next official period would be 1991 - 2020. These fixed time intervals allow world-wide comparison of climatological events.
- For special purposes, however, other climatic time scales are sometimes used.

Rainfall statistics and how they are calculated
Rainfall Estimation

The amount of rainfall can be quantified with the following characteristics:

- *Duration (h)* – Length of time over which rainfall (storm event) occurs;
- *Depth (mm)* – Total amount of rainfall occurring during the storm duration;
- *Intensity (mm/h)* – Rate of rainfall or depth of rainfall divided by the duration
- The frequency of a rainfall event is the recurrence interval of storms having the same duration and volume (depth). This can be expressed either in terms of *exceedance probability* or *return period*.
- *Exceedance Probability* – Probability that a storm event having the specified duration and volume will be exceeded in one given time period, typically 1-year.
- *Return Period* – Average length of time between events that have the same duration and volume
- Dry Intervals denote the intervals with zero rainfall or no rain.
- Rainfall variability and seasonality
- Onset, cessation and duration of the rainy season
- Wet days, wet spells and their probabilities

Rainfall and Crop water requirement averages

- Rainfall and crop water satisfaction index
- Decadal rainfall
- Decadal water requirement satisfaction index
- Heptadal rainfall
- Heptadal water requirement satisfaction index

Computation of precipitation over an area

There are three methods that an area-wide precipitation can be calculated:

- Arithmetic average: applied to moderate relief, cyclonic storm, and where precipitation gradients are not strong.
- Thiessen polygon weighted average: the gauge distribution is not even or uniform, and where precipitation gradients are strong.
- Isohyetal method: strong precipitation gradients caused by topography or by thunderstorm cells.

Estimating missing rainfall data

- The method(s) chosen depend upon the nature of the available data and the purpose of the investigation
- Regression and correlation of measurements from the station of interest with data from nearby gauges when they are functioning.

- $$Pa = 1/3 \left(N_a/N_b \times P_b + N_a/N_c \times P_c + N_a/N_d \times P_d\right)$$

Where,

P: precipitation

N: long-term normal precipitation (annual, seasonal, month)

Checking the consistency of precipitation records

- In order to maintain a homogeneous record, the precipitation record should be adjusted to eliminate the effects of instrument changes.

- Double-mass curves: the accumulated average is plotted against the accumulated annual total at station.

Measurement of Precipitation

- A rain gauge is used to measure rainfall at the point it is installed.

Why do we need to measure rainfall?

- Agriculture – what to plant in certain areas, where and when to plant, when to harvest
- Horticulture/viticulture - how and when to irrigate
- Engineers - to design structures for runoff control i.e. storm-water drains, bridges etc.
- Scientists - hydrological modelling of catchments

Things to know about rainfall

- Rainfall sites in the area of interest
- Seasonal rainfall trends
- Variability of rainfall across the state and between towns
- Area specific rainfall
- Site specific rainfall
  - Methods of measuring rainfall
  - Where to place rainfall gauges - depends on the site
  - How many gauges to place in area - depends on the experiment
  - What to do with the rainfall data

Recording Rain gauge

- Measures the exact duration of the rainfall
- When it started
- When it ended
- Measures the amount of rainfall
- The number of rainfall events
- And allows for the calculation of intensity.

  - A horizontal line means zero or no rainfall
  - Falling vertical line shows siphoning period. During siphoning period rainfall is not recorded but this is relatively small since siphoning is in a few seconds.
  - Uniform slope means uniform intensity
  - Varying slope means varying intensity

SOIL-PLANT-WATER RELATIONSHIPS

- The process of photosynthesis involves the combination of molecules of water and carbon dioxide. Hydrogen, which is dissociated from the water, is then used to reduce carbon dioxide to form a carbohydrate. Sunlight is necessary to provide the energy for splitting molecules of water into hydrogen and oxygen. The photosynthetic reaction only takes place in the presence of chlorophyll which acts as a catalyst, converting light energy into chemical energy.

- $\mathbf{6CO_2 + 6H_2O}$ radiant energy from sun 48×10^4 joules in the presence of chlorophyll $\mathbf{C_6H_{12}O_6 + 6O_2}$

Effective rainfall

Effective rainfall (P_e) is defined differently by different professionals and it depends on:

- Rainfall characteristics
- Land slope
- Characteristics of the soil
- Groundwater characteristics
- Management practices
- Crop characteristics.

Water in the Soil/Water Use

- **Field capacity** (FC): Water content in the soil 1-3 days after water has been applied and drainage has largely ceased i.e. at about ⅓ bar. FC is lower in lighter soils than in heavier soils, ranging roughly from 5 up to 25-30 %.

- **Permanent Wilting Point** (PWP): Extraction of water through evapotranspiration (ET) causes an increase of surface tension from ⅓ to 15 bars. At 15 bars plants cannot extract water and die permanently and addition of water does not revive them. PWP is easily evaluated using sunflowers. It is 2-3 % for light soils and about 20 % for heavy soils.

- **Consumptive use** (CU) is used to designate the losses due to ET and water that is used for metabolic activities. However, water used for metabolic activities is << 1 % of ET) therefore, CU≈ET.

Crop Water Requirement

It is the depth of water needed to meet the water loss through ET_C of a disease-free crop, growing in large fields under non-restricting soil conditions including soil water and fertility and achieving full production potential under the given growing environment. It is determined in 3 steps:

- The effect of climate on crop water requirements
- The effect of the crop characteristics on crop water requirements

  $\mathbf{ET_c = k_c ET_o}$
- The effect of local conditions and agricultural practices on crop water requirements.

Estimation of Evapotranspiration

1. Hydrological methods (i.e. water balance)
2. Direct measurements (i.e. lysimeters)
3. Micrometeorological (i.e. energy balance)
4. Empirical or combination methods (i.e. based on energy balance or climatic factors)
5. Satellite remote sensing for whole catchments instead of point estimates in 1-4.

Thank You

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